Data-efficient surrogates for vocal-fold models: a comparison of low-N regression strategies

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Abstract

Physically based vocal-fold models predict phonatory outputs from muscle activations and subglottal pressure but are expensive to evaluate, so machine-learning surrogates are used in their place. Surrogate accuracy, however, collapses when only a handful of simulations are available. Low sample use cases are especially relevant in subject-specific clinical use, where 10 to 100 target samples may be all that can be obtained. Transfer learning was the natural remedy, training a surrogate on a cheap source model (a body-cover model, BCM) and adapt it to an expensive target with a few target simulations. This has been shown to work between closely related lumped models, for example BCM to Triangular Body Cover Model (TBCM). Here we ask whether it extends to a higher-fidelity target: a structurally distinct finite-element Beam and Membrane (BM) model. It does not. Used on BM without adaptation, the source component of the transfer learning adds nothing. We show that a no-source alternative closes this gap. TabPFN, a tabular transformer pretrained once and then frozen, fits a task in-context with no source model and no training; it is the strongest method at low N on the higher-fidelity (BM) target and reconstructs the full $(a_{CT} \times a_{TA})$ activation map from a small number of samples. On the aligned (TBCM) target, where the cheap source is already a good prior, optimized transfer leads at the smallest sample sizes and TabPFN overtakes it from $N = 100$. For high-fidelity, low-data surrogate modeling, a no-source in-context model is thus the approach that carries over where transfer does not. We limit our conclusions to the methods, targets, and forward setting studied.

1 Introduction

Physically based models of the vocal folds, such as lumped body-cover models and higher-fidelity continuum (finite-element) models, predict phonatory outputs from muscle activations and subglottal pressure, but are costly to evaluate at scale. Prior work showed that machine-learning surrogates can replace these simulations for the forward problem, yet their accuracy degrades sharply when few training simulations are available. That degradation motivated transfer learning: train a surrogate on a cheap source model (the body-cover model, BCM) and adapt it to an expensive target with a small number of target simulations. In practice, obtaining data from complicated models is costly, so being able to circumvent that with a machine learning surrogate is ideal. This transfer has been shown to succeed between closely related lumped models (BCM to TBCM). [TODO: insert citation here, to add eventually]

That prior result raises the question this paper investigates: does the transfer-learning approach extend from closely related lumped models to a higher-fidelity target? A cheap source helps only while it remains a good prior for the target, and a finite-element model is both more detailed than the lumped source and structurally unlike it, so whether the established approach carries over is not obvious. Our central question, and its fallback, is therefore:

> Does data-efficient, subject-specific surrogate modeling, shown to transfer between lumped vocal-fold models (BCM to TBCM), extend to a higher-fidelity target (BCM to BM)? And where transfer does not carry over, can a no-source pretrained model serve in its place?

We approach this in two steps. First (data efficiency, Sec. 3.1), we measure three regressor families, polynomial, random forest, and neural network, each fit two ways: on the target data alone, to establish a baseline, and with optimized transfer learning from the source. Second (fidelity extension, Sec. 3.2), we test whether the transfer benefit that holds on the aligned lumped target survives the move to the structurally distinct, higher-fidelity BM target. Throughout we also evaluate TabPFN [1], a transformer pretrained once on millions of synthetic tabular problems and then frozen, which fits a new task in-context in a single forward pass with no per-task training and no source model. We let the results, not the premise, identify the strongest method, and we bound our conclusions to the methods and targets evaluated.

2 Methods

2.1 Physical vocal-fold models and data

All models share the inputs (cricothyroid, thyroarytenoid, and lateral cricoarytenoid activations a_{CT}, a_{TA}, a_{LCA}, and subglottal pressure P_s) and the base outputs F_0 and SPL; the richer physiological outputs (Sec. 3.3) are added for the targets. The three models span a deliberate range of source–target fidelity:

- **Body-Cover Model (BCM), source.** A lumped-element model and the cheap source ($\sim$360,750 samples) for all transfer methods.
- **Triangular Body-Cover Model (TBCM), aligned target.** Lumped-element, the same modeling family as BCM, so the source–target gap is small. Sampled on a 40×40 ($a_{CT} \times a_{TA}$) grid, providing dense ODE ground-truth activation maps (Sec. 3.2). This is the near target for Q2.
- **Beam and Membrane (BM), far target.** A finite-element and continuum model; the expensive, structurally distinct target. Relative to the lumped BCM source it is the far target, where the low-N question is sharpest and the fidelity gap is largest. 4,647 phonating samples (filtered ACFL > 30).

The BCM to BM gap is not merely one of accuracy but of representation: BM resolves distributed tissue displacement fields where BCM lumps them into a few masses, so quantities such as AC flow can differ in definition and scale between the two (Sec. 3.3). [TODO: add model equations and refs; cite Parra TBCM and the BM solver; state the BCM to BM structural differences precisely.]

2.2 Low-N regression strategies

We evaluate three standard regressor families, each fit two ways, plus a pretrained no-source model, giving seven configurations under one protocol.

Regressors fit on the target alone (baseline). To measure what transfer learning actually contributes, we first fit each regressor family on the target data by itself, with no source model. This is not a proposed method but a reference point: if a transfer method cannot beat the same regressor trained on the target data alone, the source has added nothing. Every transfer and no-source result below is read against this reference.

- **Polynomial (PR).** A degree-3 polynomial regressor fit on the N target rows, on standardised inputs, with a ridge penalty chosen per target by leave-one-out generalised cross-validation over $\alpha \in [10^{-3}, 10^4]$.
- **Random forest (RF).** A random forest fit on the N target rows: 300 trees throughout, with depth and leaf size scaled to the sample count ($N < 50$: depth 3, min leaf 1; $N < 250$: depth 6, min leaf 2; $N < 1000$: depth 10, min leaf 2; $N \geq 1000$: depth unlimited, min leaf 1).
- **Neural network (NN).** A feed-forward network with two hidden layers of 64 ReLU units and one linear output per target, trained on the target sample full-batch with Adam (learning rate 0.003, weight decay 0.0001) for 800 epochs on standardised inputs and outputs.

Optimized transfer learning (per family). Each family also appears as an optimized transfer method: the source model (per-domain pretraining on BCM) is combined with a sub-model fit on the target alone, a residual-correction sub-model, and a feature-augmentation sub-model, blended by non-negative weights fit on the target sample. The same procedure is applied to all three families. The blend weights are fit on out-of-fold sub-model predictions ($K = \min(5, N)$-fold) rather than on the training rows themselves. The neural-network source model is pretrained on a 20,000-row subsample of the source for 300 mini-batch epochs at learning rate 0.001.

TabPFN (no source). A pretrained tabular transformer applied with per-output standardization, one single-output head per output. No training beyond the in-context fit, and no source model.

Evaluation. For each training size $N \in \{10, 20, 50, 100, 200, 500\}$, every method trains on N target rows and is evaluated on a disjoint held-out pool of up to 1000 rows, reporting mean $\pm$ standard deviation over 5 seeds. We report accuracy (R^2) and a range-normalized RMSE,

$$\mathrm{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}}{\max_i y_i - \min_i y_i}, \tag{1}$$

rather than MAPE, because SPL, collision pressure, and CPP cross zero, which makes a percentage error ill-defined. All methods share the same splits and seeds, so differences reflect the method rather than the data draw.

3 Results

3.1 Q1, data efficiency: accuracy versus training size

On the far (BM) target, TabPFN attains the highest R^2 and the lowest normalized RMSE from $N = 50$ upward, with the largest margins in the low-N regime (Fig. 1, Table 1). The cheap BCM source is a poor prior for the finite-element BM model. Used on BM without adaptation it is less accurate than simply predicting each output's average value, so the transfer methods have nothing useful to build on and gain little over fitting the same regressor on the target data alone: their R^2 values nearly coincide at every training size (Table 1). TabPFN, which needs no source at all, is ahead of both. The richer physiological outputs (Sec. 3.3) follow the same ordering. As N grows the methods converge toward high accuracy, yet TabPFN remains in front.

Table 1: Q1. Accuracy (R^2, upper block) and range-normalised RMSE (lower block) on the far BM target at $N = 50$ target simulations, mean over 5 seeds. Best per row in bold. The *target alone* columns are each regressor family fit on the N target rows with no source; *optimized transfer* adds the BCM source through the blended procedure of Sec. 2.2. At $N = 200$, F_0 reaches 0.75 (RF alone), 0.84 (RF transfer), and 0.92 (TabPFN).

	Fit on target alone			Optimized transfer			No source
Output	PR	RF	NN	PR	RF	NN	TabPFN
R^2 *(higher is better)*							
F0	0.85	0.60	0.79	0.85	0.77	0.85	**0.89**
SPL	0.72	0.69	0.68	0.72	0.67	0.72	**0.81**
ACFL	0.89	0.78	0.85	0.89	0.79	0.87	**0.92**
PC	0.57	0.60	0.36	0.61	0.58	0.47	**0.71**
CPP	0.36	0.40	-0.04	0.41	0.38	0.03	**0.47**
nRMSE (lower is better)							
F0	0.09	0.14	0.10	0.09	0.11	0.09	**0.08**
SPL	0.07	0.07	0.07	0.07	0.07	0.06	**0.05**
ACFL	0.07	0.09	0.08	0.07	0.09	0.07	**0.06**
PC	0.15	0.14	0.18	0.14	0.15	0.16	**0.12**
CPP	0.16	0.15	0.20	0.15	0.16	0.19	**0.15**

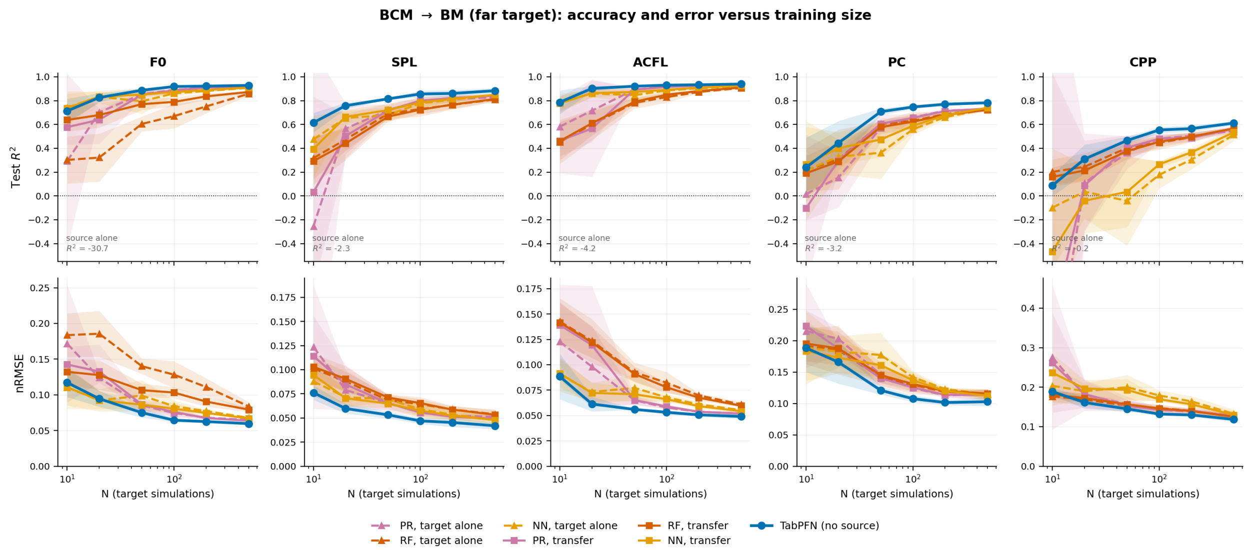

Figure 1: **Q1, data efficiency.** Accuracy and error versus training size on the far BM target (all five outputs): R^2 (top) and normalized RMSE (bottom) versus N for the regressors fit on the target alone (dashed, our baseline), the optimized transfer methods (solid), and TabPFN (thick blue), mean $\pm$ s.d. over 5 seeds. The source-alone R^2, annotated per panel, falls far below the axis and illustrates the poor BCM prior on this target.

3.2 Q2, model complexity: when does transfer help?

Whether the source helps at all is judged against the baseline, each regressor fit on the target data alone. On the aligned lumped target (TBCM, the same family as the BCM source), the source is already a strong prior, by itself reaching an F_0 R^2 of 0.86 with no target data, and optimized transfer clears the baseline, raising the F_0 R^2 from 0.94 to 0.97 at $N = 50$. On the far continuum target (BM), the same source is less accurate than guessing each output's average value, and transfer does not clear the baseline at all, an F_0 R^2 of 0.854 against 0.851: the source is mostly dead weight. Transfer's payoff therefore tracks how close the source is to the target, exactly the principle it rests on. TabPFN, which uses no source, is the strongest method on the far target and stays close to the best on the aligned one, with an F_0 R^2 of 0.96 on TBCM and 0.89 on BM at $N = 50$, and so removes the dependence on a well-aligned source altogether.

Table 2: Q2. Source–target fidelity. F_0 accuracy (R^2) at $N = 50$ as the target moves from an aligned lumped model (TBCM) to a structurally distinct continuum model (BM). *Baseline* is the best of the three regressor families fit on the target alone; *opt. transfer* is the best of the three transfer variants. Transfer is only worth its source when it beats the baseline column. A negative source-alone value means the unadapted BCM source is less accurate than predicting the mean output. Transfer clears the baseline when aligned and not at all when far; TabPFN needs no source and wins on the far target.

Target (fidelity)	Source alone	Baseline	Opt. transfer	TabPFN
TBCM (aligned lumped)	0.86	0.94	**0.97**	0.96
BM (far continuum)	-30.9	0.85	0.85	**0.89**

Beyond point accuracy, a useful surrogate must reconstruct the physiological structure of the

($a_{CT} \times a_{TA}$) control space. On TBCM, which carries a dense ODE ground-truth map, TabPFN reconstructs the full motor-control surface from as few as 25 samples and converges to the ground truth by $N = 1000$, with lower image-level error than optimized transfer from $N = 50$ upward (Fig. 2). The same behavior holds on BM, where against a dense full-data reference TabPFN resolves the correct surface by $N \approx 25$ while the transfer surface stays blocky and dragged toward the misaligned source.

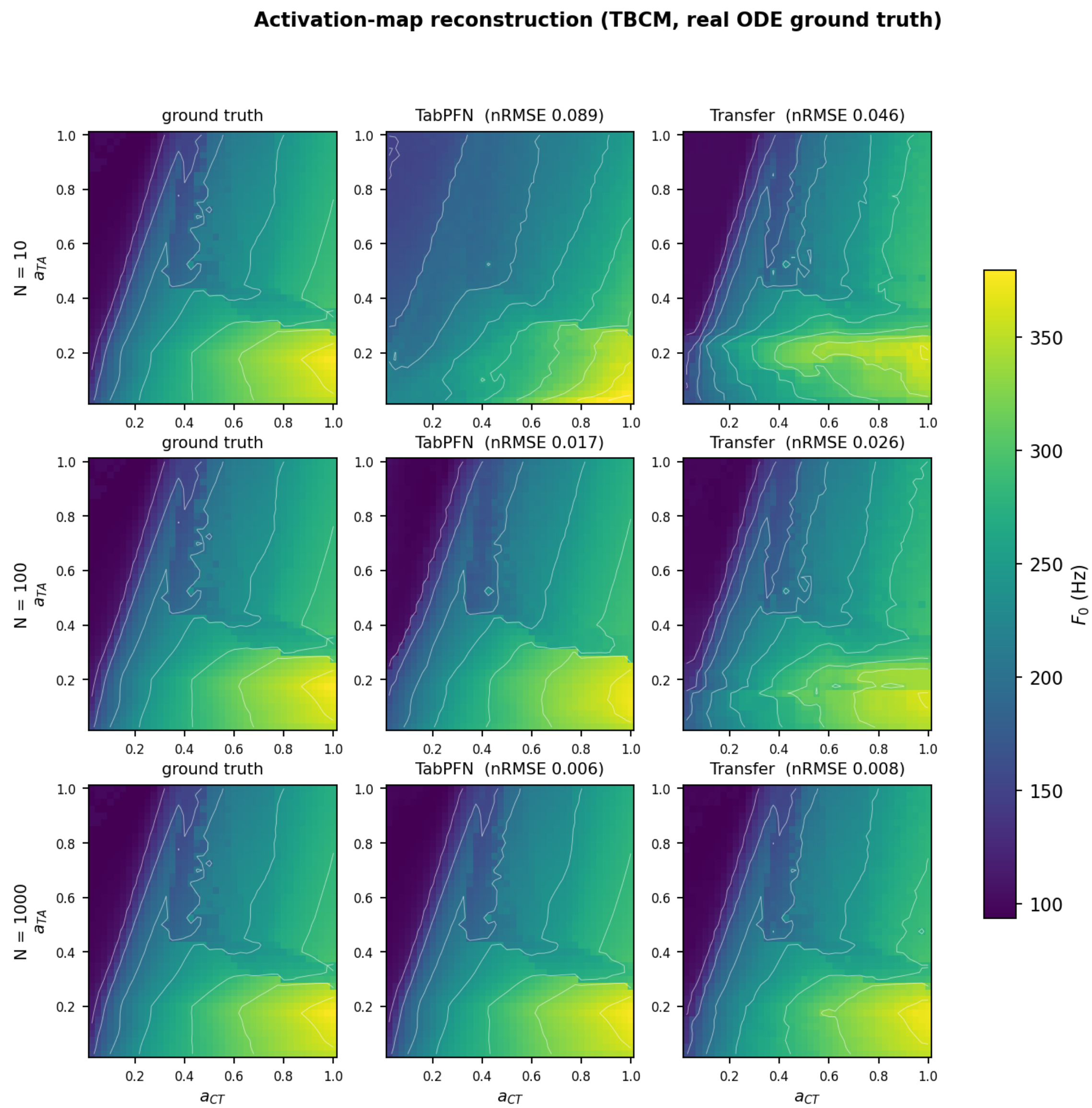

Figure 2: **Activation-map reconstruction (TBCM, real ODE ground truth).** F_0 motor-control map over (a_{CT}, a_{TA}). Columns: ODE ground truth, TabPFN, optimized transfer; rows: increasing N (10/100/1000). Image-level nRMSE over all 1600 cells is given per panel. TabPFN recovers the correct structure from small samples; the transfer surface is distorted at low N and sharpens only as N grows.

3.3 Richer physiological outputs

F_0 and SPL are the conventional surrogate targets, but clinical use turns on quantities describing tissue loading and voice quality. The BM target therefore carries three further outputs: AC glottal flow (ACFL), peak collision pressure (PC), and cepstral peak prominence (CPP), regenerated row-aligned to the original operating points so that they correspond to the same simulations as F_0 and SPL. The ordering of methods is unchanged on these outputs (Table 1). What changes is the difficulty: ACFL is the easiest for every method (TabPFN $R^2 = 0.92$ at $N = 50$), CPP by a clear margin the hardest ($R^2 = 0.47$), with PC intermediate ($R^2 = 0.71$). CPP is a cepstral voice-quality measure computed from the radiated pressure waveform rather than a direct mechanical quantity, and is the output least determined by the three input activations alone, which is the most likely reason no method reaches high accuracy on it at any N tested. [TODO: ACFlow still provisional.]

4 Discussion

The two questions resolve into one picture. On data efficiency, a pretrained tabular transformer is the strongest low-N surrogate among the methods tested on the far target, and is competitive on the aligned one. On model complexity, the benefit of transfer learning is governed by how close the source is to the target: it is large when the target is close to the cheap source (TBCM) and negligible when the target is a structurally distinct, higher-fidelity model (BM). Because TabPFN needs no source model or source data, it is insensitive to that fidelity gap, needing neither a well-aligned source nor the per-domain pretraining a source demands.

We state these conclusions narrowly. They hold for the seven configurations evaluated, the two targets studied (TBCM and BM), and the forward problem; they do not establish that TabPFN is optimal in general, nor do they speak to methods we did not test (for example, physics-informed networks) or to targets and output sets we did not examine.

Transfer learning for a no-source model (future work). Because TabPFN wins at low N but has no weights to fine-tune, giving it a form of transfer is an open direction: rather than weight transfer, a linear combination of source-fit and target-fit in-context models may import source structure while preserving the no-training property. [TODO: report feasibility once explored.]

A property under missing data (future work). A pretrained in-context model can also be queried when parts of the input space are unobserved. In preliminary experiments, TabPFN reconstructs held-out regions of the control space and degrades gracefully as the missing fraction grows. Because this is not straightforwardly comparable across all methods, we note it here as a direction rather than a result.

5 Conclusion

Across a controlled comparison of three regressor families (each fit on the target alone and with transfer) and a pretrained tabular transformer, on targets spanning a range of source fidelity, two findings hold. First, TabPFN is the most accurate low-N surrogate on the far target, no source and no training. Second, the payoff of transfer learning is set by how close the source is to the target, substantial for a closely related lumped target and negligible for a far continuum target. A no-source in-context model sidesteps that dependence. We present this as a finding about the methods and targets evaluated, not as a universal claim.

References

[1] N. Hollmann, S. Müller, K. Eggensperger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in *International Conference on Learning Representations (ICLR)*, 2023.